

Charles Darwin University
PRT661 Data Science Practice Unit

Assessment 1

Project Proposal and Design

Solar Generation Forecasting for Battery Dispatch

DKASC Alice Springs · 2008–2026

Theme 2: Predictive Analytics and Forecasting

Group 4

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Repository: github.com/Uttam1618/PRT661-solar-forecasting

Jira board: [SCRUM — PRT661 Solar Forecasting Group 4](#)

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1. Project overview and objectives

Group 4 will predict photovoltaic generation at the Desert Knowledge Australia Solar Centre (DKASC) site in Alice Springs, 1 to 24 hours in advance, to inform a site owner or operator on when a battery should charge or discharge. Forecast and uncertainty are the deliverable, it's not a system that we build, but the scheduling of the battery is the operational use it informs. Four goals are identified: create a batch pipeline that can be reproduced for nineteen yearly extracts (2008-2026) containing some 1.9 million rows and 3.1 GB of data; develop degradation-aware and ramp-sensitive features; evaluate proposed models against a published benchmark for this site; and display forecasts and forecast errors in a dashboard. It has a design that covers all five of the capabilities: acquisition, storage, processing, analytics/machine learning and visualisation.

2. Theme justification

The project is located in Theme 2, Predictive Analytics and Forecasting, which predicts a continuous future outcome based on historical data, with time-series forecasting being explicitly mentioned in the examples in the theme. Theme 1 was rejected, since dispatch decision does not need to be done with the minimum latency, so it is a batch pipeline and not a streaming one. The reason for excluding Theme 3 is because per-inverter anomaly channels are out of scope, and the fifty or so per-array columns are not included in the feature set for this reason.

3. Problem statement

Solar generation at Alice Springs is dependent on the presence of cloud and on the season, plus a slow degradation of the solar panels as they age. A battery only pays for itself if the operator can predict what generation will be available, and this battery energy storage system was installed in 2024 at the site. Thuseethan et al. (2025) predict this same location, but accuracy is an objective, they use 2019 to 2024, and they pre-date the battery. Three gaps follow. The 2024 installation provides dispatchable SoC (State of Charge) signals. The 17-year period shows degradation not apparent in a 5-year period. The previous studies reported high error at dawn/dusk and peak irradiance but did not investigate what aspects cause the high error. This project is designed to forecast to support operational decision and directly tackles ramp events and feature importance. We are going to use their best published result - R^2 0.8641 and MSE 22.41 on Master Meter 2 to measure up against.

4. Initial architecture

PRT661 Solar Generation Forecasting High-Level System Architecture

DKASC Alice Springs · 2008–2026 · Theme 2 Predictive Analytics and Forecasting

Objective: forecast generation 1–24 hours ahead so that battery charge and discharge can be scheduled against expected output

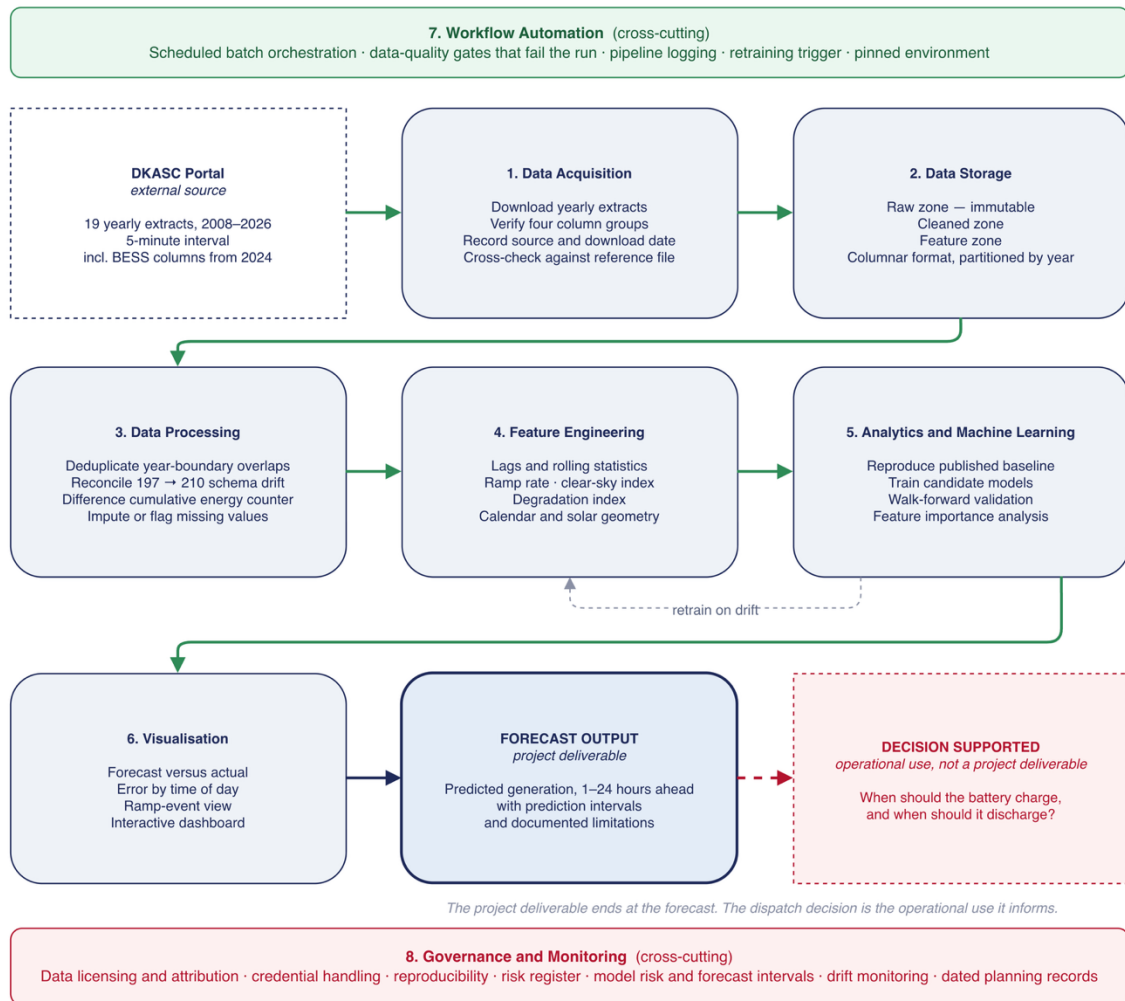


Figure 1. High-level system architecture. The forecast output is computed independently of the dispatch decision that it provides.

The high-level design is shown in Figure 1. The data follows six stages from the DKASC portal to a forecast, with workflow automation and governance spanning all six stages. The forecast is distinct from the decision, which is the edge of what this project will provide. The storage uses parquet files queried through DuckDB; PostgreSQL was rejected as at 1.9 million rows, the overhead would not be justified, and Apache Spark as the working set would be in memory. These both add to the cost of operation, but do not enhance the results. All components are open source and can be reproduced from the repository and Figure 2 shows the design, layer-by-layer.

Data Pipeline Architecture — Layered View

Batch pipeline · scheduled execution · all open-source components



Figure 2. Data pipeline architecture layered view: component, output format and rebuild cadence at each layer.

5. Workflow plan

Workflow Plan — Epics, Owners and Milestones

Every epic name below appears identically as a Jira epic in the project board



Figure 3. Workflow plan: the eight epics against the four assessment milestones.

The following eight epics are shown against the four assessment milestones in figure 3. The eight names of epic in figure 3 are the eight epic words on the Jira board, and they are worded exactly the same, and each epic is assigned to its accountable owner, as a Jira assignee. The board has 18 tasks,

each of which has been assigned to a member of the team and to a sprint, seven of which are Done, four In Progress and seven To Do, across four sprints corresponding to the A1, A2, A3 and A4 work areas. Progress, scope changes and completion status are recorded as work takes place and not regenerated when it is submitted.

6. Task allocation and team structure

Group 4 has four members. Table 1 provides a list of roles and epic ownership: each member owns two epics across the entire pipeline order with the Project Lead also having named tasks within Feature Engineering and Analytics and Machine Learning as second member, ensuring that each epic has two members who are familiar with it. Work is done when it is committed, reviewed by another member and their Jira task closed with an evidence link. Decisions are made documented on dated planning records and not agreed verbally.

Role	Member	Epics (accountable)	Report sections
Project Lead	Uttam Shrestha	Workflow Automation, Governance and Monitoring	5, 6, 9 and assembly
Data Engineer	Abhishek Tamang	Data Acquisition, Data Storage	1, 3
Data and Feature Engineer	Sarin Uprety	Data Processing, Feature Engineering	4, 7
Modelling and Visualisation Lead	Yogesh Basnet	Analytics and Machine Learning, Visualisation	2, 8

Table 1. Roles, epic ownership and report sections.

7. Initial risk analysis

The repository's risk register lists 19 risks under the categories of technical, project and governance, including likelihood, impact, mitigation and a risk owner. The seven most significant, by impact, are shown in Table 2. The register is checked at each stand up meeting and each submission; review of 12 August resulted in giving owners to named members and thus matching each risk to a named Jira assignee. Just two risks have been realised and mitigated; the first being the slippage against the internal draft deadline and the second being a stale extract which ends 11 February 2026.

ID	Risk	Impact	Mitigation
T1	Schema drift: 197 columns to 2023, 210 from 2024	Medium	The pipeline takes care to reconcile both schemas explicitly, indicating that the BESS columns are only available from 2024 onwards.
T3	Cumulative counter is giving the wrong readings as interval energy.	High	The cumulative series is differenced, checked for validity with the reference extract and unit-tested prior to use.
T4	High missingness: wind speed 52%, tilted radiation ~35%	Medium	Wind speed is removed and the other missing values are imputed or flagged, and it is noted which of these was done.

ID	Risk	Impact	Mitigation
T5	Stale extract: 2026 data ending on 11th February 2026	Medium	Before A2, the data is re-downloaded and the download date is noted; and A1 is truthful about its coverage.
T8	Overfitting through inappropriate validation	High	No random shuffling at any time, instead it is walking forward validation.
P1	Uneven contribution across the group	High	All work has a named Jira assignee, is discussed weekly in the stand-up and issues are escalated upfront rather than on submission.
G4	Poor dispatch decision due to inaccurate forecast.	Medium	Prediction intervals are reported for forecasts and known weak periods are openly reported as limitations.

Table 2. Extract from the initial risk register, rated by impact.

8. Ethics, privacy and security

No personal information is included in the dataset. It simply documents electrical data from a demonstration facility and data from a co-located weather station, and no one can be identified in either case, thus no consent process is necessary. Data here and in repository is credited to DKASC, and source URL and date of download recorded to allow results to be traced back to a specific extract; raw files are not version controlled.

Seasonal calendars of First Nations were used in previous work on this site. We intentionally do not emulate that approach. This is because it is the substantive work of the authors, and Indigenous seasonal knowledge is cultural intellectual property and should only be used following engagement with Traditional Owners, which is beyond the scope and timeframe of this unit.

There is an operational cost to a poor forecast – not just a statistical cost. Prediction intervals are thus reported along with point forecasts, known weak time periods are recorded, and the system is stated to be a decision support system, not an automated controller. Credentials are never committed: .gitignore excludes environment, key and credential files. Where generative AI may be used to aid in drafting or coding, its use is identified in accordance with CDU policy.

9. References

- Data download | DKA solar centre. (n.d.). In *dkasolarcentre.com.au*. Retrieved 16 August 2026, from <https://dkasolarcentre.com.au/download?location=alice-springs>
- Thuseethan, S., Gangajaliya, S., Hamlin, L., Shanmugam, B., & Thennadil, S. (2025). Conv-ensemble for solar power prediction with first nations seasonal information. *IEEE Open Journal of the Computer Society*, 6, 884–895. <https://doi.org/10.1109/OJCS.2025.3580339>